Chapter 4. Brownian Motion

Definition and path properties, Lévy's construction, the Markov property, d-dimensional Brownian motion

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4.1 Definition and some properties

Motivation. Brownian motion is the continuous-time limit of the random walk of Example 3.8: Gaussian, independent increments and continuous paths. This section fixes the definition, records its two basic symmetries, and shows that a typical path has quadratic variation t and therefore infinite length on every interval, so it cannot be treated by classical calculus.

Let $(\Omega, \mathcal{F}, P)$ be a probability space and $I \subset \mathbb{R}_+ := [0, \infty)$ an index set, e.g. $\{0, \dots, n\}$, $\mathbb{N}$, $\mathbb{N}_0$, $[0,1]$, $[0,T]$, $[0,T)$, the positive dyadic rationals $\mathbb{Q}_2^+ := \{m2^{-n} \mid m, n \geq 0\}$, or $[0,\infty)$. A **stochastic process** is a family $(X_t)_{t\in I}$ of r.v.'s on $(\Omega, \mathcal{F}, P)$; for each ω, $t \mapsto X_t(\omega)$ is a **sample path**. More generally X_t may take values in a metric space S (the **state space**), measurably w.r.t. $\mathcal{B}(S)$. We read t as time and write $(X_t)_{t\geq 0}$ when $I = \mathbb{R}_+$.

Definition 4.1 (Brownian motion)

A real-valued process $(B_t)_{t\geq 0}$ is a (standard) **Brownian motion** (BM) if

(i) $P(B_0 = 0) = 1$;

(ii) for $0 \leq s < t$, $B_t - B_s \sim N(0, t-s)$, i.e. $P(B_t - B_s \leq b) = \frac{1}{\sqrt{2\pi(t-s)}} \int_{-\infty}^{b} e^{-x^2/(2(t-s))} dx$;

(iii) it has **independent increments**: for $0 \leq t_1 < \cdots < t_n$, the r.v.'s $B_{t_1} - B_0, B_{t_2} - B_{t_1}, \dots, B_{t_n} - B_{t_{n-1}}$ are independent;

(iv) almost all paths are continuous: $P(t \mapsto B_t \text{ is continuous on } [0,\infty)) = 1$.

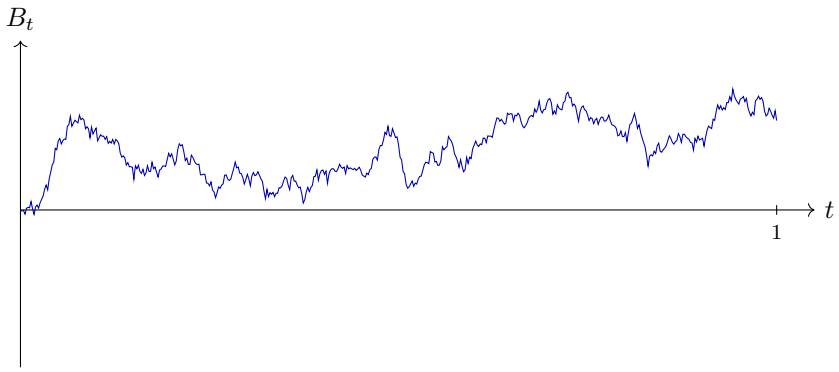


Figure. *A simulated Brownian path on $[0,1]$ (2000 Gaussian increments of variance $1/2000$). It is continuous but wiggles at every scale: its quadratic variation over $[0,t]$ is t (Theorem 4.3), so its total variation is infinite (Proposition 4.6).*

Proposition 4.2

Let $(B_t)_{t\geq 0}$ be a BM. (i) (Translation invariance) For fixed $t_0 \geq 0$, $\tilde{B}_t := B_{t+t_0} - B_{t_0}$ is a BM. (ii) (Scaling invariance) For $\lambda > 0$, $\tilde{B}_t := \frac{1}{\sqrt{\lambda}} B_{\lambda t}$ is a BM.

Proof. (i) Definition 4.1(i),(iv) are clear, and $\tilde{B}_t - \tilde{B}_s = B_{t+t_0} - B_{s+t_0} \sim N(0, t-s)$. For $0 \leq t_1 < \cdots < t_n$, the increments of $\tilde{B}$ are increments of B along $t_0 \leq t_1 + t_0 < \cdots < t_n + t_0$ (if $t_1 = 0$ the first one is the constant 0, which is independent of everything), hence independent by (iii) for B.

(ii) (i), (iii),

(iv) carry over directly. For (ii), substituting $y = x/\sqrt{\lambda}$,

$$P(\tilde{B}_t - \tilde{B}_s \leq b) = P(B_{\lambda t} - B_{\lambda s} \leq \sqrt{\lambda} b) = \int_{-\infty}^{\sqrt{\lambda} b} \frac{e^{-x^2/(2\lambda(t-s))}}{\sqrt{2\pi\lambda(t-s)}} dx = \int_{-\infty}^{b} \frac{e^{-y^2/(2(t-s))}}{\sqrt{2\pi(t-s)}} dy.$$

■

Let $\tau_n = \{0 = t_1^{(n)} < t_2^{(n)} < \cdots < t_{N(\tau_n)}^{(n)} < \infty\}$ be partitions of $[0,\infty)$ with

$$|\tau_n| := \max_i \left(t_{i+1}^{(n)} - t_i^{(n)}\right) \to 0 \quad \text{(mesh)}, \qquad t_{N(\tau_n)}^{(n)} \to \infty. \tag{$\star$}$$

Theorem 4.3 (Quadratic variation)

Let (B_t) be a BM and (τ_n) as in $(\star)$.

(i) For every $t \geq 0$, $S_t^n := \sum_{t_i^{(n)} \in \tau_n,\ t_{i+1}^{(n)} \leq t} \left(B_{t_{i+1}^{(n)}} - B_{t_i^{(n)}}\right)^2 \to t$ in $L^2(\Omega, \mathcal{F}, P)$.

(ii) If $\tau_n \subset \tau_{n+1}$ for all n, or $\sum_n |\tau_n| < \infty$, then $P(\lim_n S_t^n = t \ \forall t \geq 0) = 1$.

Proof. (i) Write $\Delta_i := t_{i+1}^{(n)} - t_i^{(n)}$; the sums run over i with $t_{i+1}^{(n)} \leq t$. Then $\mathbb{E}[S_t^n] = \sum \Delta_i \to t$ in $\mathbb{R}$, by $(\star)$. By independence of increments and $\mathbb{E}[N^4] = 3\sigma^4$ for $N \sim N(0,\sigma^2)$,

$$\operatorname{Var}(S_t^n) = \sum \operatorname{Var}\left((B_{t_{i+1}} - B_{t_i})^2\right) = \sum (3\Delta_i^2 - \Delta_i^2) = 2\sum \Delta_i^2 \leq 2t|\tau_n| \to 0.$$

Hence $S_t^n - \mathbb{E}[S_t^n] \to 0$ in L^2, and with $\mathbb{E}[S_t^n] \to t$ we get $S_t^n \to t$ in L^2.

(ii) [Supplement] *Case* $\sum_n |\tau_n| < \infty$ (this includes the dyadic partitions $t_i^{(n)} = (i-1)2^{-n}$, the case treated in the lectures).

Step 1 (fixed t). By Chebyshev and the bound above, $\sum_n P(|S_t^n - \mathbb{E}S_t^n| \geq \varepsilon) \leq \varepsilon^{-2} \sum_n 2t|\tau_n| < \infty$. By Lemma 2.10, $S_t^n - \mathbb{E}[S_t^n] \to 0$ a.s., hence $S_t^n \to t$ on a set Ω_t with $P(\Omega_t) = 1$.

Step 2 (all t at once). Let $\Omega' := \bigcap_{t \in \mathbb{Q}_2^+} \Omega_t$, so $P(\Omega') = 1$. For $\omega \in \Omega'$ and arbitrary t, take dyadic $t_k^- \nearrow t$, $t_k^+ \searrow t$. Since S_t^n is increasing in t,

$$t_k^- = \lim_n S_{t_k^-}^n(\omega) \leq \liminf_n S_t^n(\omega) \leq \limsup_n S_t^n(\omega) \leq \lim_n S_{t_k^+}^n(\omega) = t_k^+,$$

and $k \to \infty$ gives the claim.

General nested partitions: see Revuz–Yor, *Continuous Martingales and Brownian Motion*, Ch. II, Prop. (2.12); not proved in the lectures. ■

Remark 4.4

Let Ω'' with $P(\Omega'') = 1$ be such that $t \mapsto B_t(\omega)$ is continuous for $\omega \in \Omega''$, and $\tilde{\Omega} := \Omega' \cap \Omega''$ with Ω' from the proof above. For $\tilde{\omega} \in \tilde{\Omega}$ the path is continuous and, along (τ_n) as in Theorem 4.3(ii),

$$\langle B \rangle_t(\tilde{\omega}) := \lim_n \sum_{t_{i+1}^{(n)} \leq t} \left(B_{t_{i+1}^{(n)}}(\tilde{\omega}) - B_{t_i^{(n)}}(\tilde{\omega})\right)^2 = t \qquad \forall t \geq 0.$$

Such a path is a **typical path**, and $\langle B \rangle_t(\tilde{\omega})$ its **quadratic variation**.

Definition 4.5 (Finite variation)

$g : \mathbb{R}_+ \to \mathbb{R}$ has **finite variation** on $[0,t]$ if

$$\operatorname{Var}_t g := \sup_\tau \sum_{t_i \in \tau,\ t_{i+1} \leq t} |g(t_{i+1}) - g(t_i)| < \infty, \tag{4.1}$$

the supremum over all partitions $0 = t_1 < \cdots < t_{N(\tau)} < \infty$; g has finite variation if this holds

for every t.

Proposition 4.6

A typical path of BM is continuous and of unbounded variation on every interval $[0,T]$, $T>0$.

Proof. Suppose $\mathrm{Var}_T B_\cdot(\tilde\omega) < \infty$ for some $T>0$. The path is uniformly continuous on $[0,T]$, so for $\varepsilon>0$ the increments are $<\varepsilon$ once $|\tau_n|<\delta$, and

$$\sum_{t_{i+1}\le T} \big(B_{t_{i+1}} - B_{t_i}\big)^2(\tilde\omega) \le \max_i |B_{t_{i+1}} - B_{t_i}|(\tilde\omega) \cdot \sum_i |B_{t_{i+1}} - B_{t_i}|(\tilde\omega) \le \varepsilon\, \mathrm{Var}_T B_\cdot(\tilde\omega).$$

Hence $\langle B\rangle_T(\tilde\omega) = 0$, contradicting $\langle B\rangle_T(\tilde\omega) = T > 0$. ∎

4.2 Construction (hence existence) of Brownian motion

Motivation. Does Brownian motion exist? Kolmogorov's extension theorem builds a process with the prescribed Gaussian finite-dimensional laws on $\mathbb{R}^I$, but the set of continuous paths is not measurable there. We therefore build the process on the dyadic times, show with the Kolmogorov–Čentsov criterion that its paths are a.s. uniformly continuous, and extend them continuously.

For a process $(X_t)_{t\in I}$ and pairwise distinct $t_1,\dots,t_n \in I$ let

$$\mu_{t_1\dots t_n}(A_1\times\cdots\times A_n) := P\big((X_{t_1},\dots,X_{t_n}) \in A_1\times\cdots\times A_n\big), \qquad A_i\in\mathcal{B}(\mathbb{R}). \tag{4.2}$$

By Theorem 1.9 this determines the law $P((X_{t_1},\dots,X_{t_n})\in A)$, $A\in\mathcal{B}(\mathbb{R}^n)$, because rectangles are a $\cap$-stable generator of $\mathcal{B}(\mathbb{R}^n)$. These **finite-dimensional distributions** are **consistent**:

$$\mu_{t_1\dots t_n}(A_1\times\cdots\times A_n) = \mu_{t_{\pi(1)}\dots t_{\pi(n)}}(A_{\pi(1)}\times\cdots\times A_{\pi(n)}) \quad \text{for permutations } \pi, \tag{4.3}$$

$$\mu_{t_1\dots t_n}(A_1\times\cdots\times A_{n-1}\times\mathbb{R}) = \mu_{t_1\dots t_{n-1}}(A_1\times\cdots\times A_{n-1}) \quad (n\ge 2). \tag{4.4}$$

Any family of probability measures $\mu_{t_1\dots t_n}$ on $\mathbb{R}^n$ satisfying (4.3), (4.4) is called consistent. Conversely, does every consistent family come from a process?

On $\mathbb{R}^I := \{\omega : I\to\mathbb{R}\}$ let the **cylinder sets** be

$$\mathcal{C}_I := \big\{\{\omega \mid (\omega(t_1),\dots,\omega(t_n))\in A_1\times\cdots\times A_n\} \;\big|\; t_1<\cdots<t_n\in I,\ A_i\in\mathcal{B}(\mathbb{R}),\ n\in\mathbb{N}\big\}.$$

$\sigma(\mathcal{C}_I)$ is the smallest σ-algebra making all finite-dimensional projections $f_{t_1\dots t_n}(\omega) := (\omega(t_1),\dots,\omega(t_n))$ measurable, and, since $f^{-1}_{t_1\dots t_n}(A_1\times\cdots\times A_n) = \bigcap_i f_{t_i}^{-1}(A_i)$, even the smallest one making all f_t, $t\in I$, measurable.

Theorem 4.7 (Kolmogorov's extension theorem)

For every consistent family $\mu_{t_1\dots t_n}$ there is a unique probability measure P on $(\mathbb{R}^I,\sigma(\mathcal{C}_I))$ such that, with $X_t(\omega):=\omega(t)$, (4.2) holds, i.e. $\mu_{t_1\dots t_n} = P\circ f^{-1}_{t_1\dots t_n}$.

Proof [Supplement].

Step 1 (inner regularity). Every probability measure μ on $(\mathbb{R}^n,\mathcal{B}(\mathbb{R}^n))$ satisfies $\mu(A)=\sup\{\mu(K)\mid K\subset A \text{ compact}\}$. The class of A such that for every ε there are F closed $\subset A\subset G$ open with $\mu(G\setminus F)<\varepsilon$ is a σ-algebra. It contains every closed F (take $G=\{d(\cdot,F)<\frac{1}{m}\}$ and continuity from above), hence all Borel sets. Intersecting F with a large closed ball gives a compact K.

Step 2 (a finitely additive P_0). The general cylinders $Z = \{\omega \mid f_J(\omega) \in A\}$, $J \subset I$ finite, $A \in \mathcal{B}(\mathbb{R}^J)$, form an algebra $\mathcal{Z}$ with $\sigma(\mathcal{Z}) = \sigma(\mathcal{C}_I)$. Put $P_0(Z) := \mu_J(A)$. This is well defined: two representations can be lifted to a common finite $J' \supset J$, and (4.3), (4.4), which extend from rectangles to all Borel sets by Theorem 1.9, show that $\mu_J(A) = \mu_{J'}(A \times \mathbb{R}^{J' \setminus J})$. Since finitely many cylinders can be written over one common J, P_0 is finitely additive with $P_0(\mathbb{R}^I) = 1$.

Step 3 (σ-additivity). By Theorem 1.15 it suffices that $Z_k \in \mathcal{Z}$, $Z_k \searrow \emptyset$ imply $P_0(Z_k) \to 0$. Suppose $P_0(Z_k) \geq \varepsilon > 0$ for all k, with $Z_k = \{f_{J_k} \in A_k\}$ and $J_1 \subset J_2 \subset \cdots$. By Step 1 choose compact $K_k \subset A_k$ with $\mu_{J_k}(A_k \setminus K_k) \leq \varepsilon 2^{-k-1}$, and put $E_k := \bigcap_{j \leq k} \{f_{J_j} \in K_j\} \subset Z_k$. Then $P_0(Z_k \setminus E_k) \leq \varepsilon/2$, so $P_0(E_k) \geq \varepsilon/2$ and $E_k \neq \emptyset$; pick $\omega_k \in E_k$. On the countable set $J := \bigcup_k J_k$, for each $t \in J_j$ the values $\omega_k(t)$, $k \geq j$, lie in the compact projection of K_j. A diagonal argument gives a subsequence with $\omega_{k_l}(t) \to \omega(t)$ for all $t \in J$; put $\omega(t) := 0$ off J. For each j, $f_{J_j}(\omega_{k_l}) \in K_j$ for $k_l \geq j$ and K_j is closed, so $f_{J_j}(\omega) \in K_j \subset A_j$. Thus $\omega \in \bigcap_j Z_j = \emptyset$, a contradiction.

Step 4. Carathéodory's Theorem 1.11 (on the algebra $\mathcal{Z}$, finite measure) extends P_0 uniquely to $\sigma(\mathcal{C}_I)$, and $P \circ f_{t_1 \dots t_n}^{-1} = \mu_{t_1 \dots t_n}$ by construction. Uniqueness: $\mathcal{Z}$ is a $\cap$-stable generator (Theorem 1.9). ■

The Brownian finite-dimensional laws. Let $I = \mathbb{R}_+$ and

$$p_t(x,y) := \frac{1}{\sqrt{2\pi t}} e^{-\frac{(y-x)^2}{2t}} \quad (t > 0), \qquad p_0(x,y)\,dy := \delta_x(dy). \tag{4.5}$$

For $0 \leq t_1 < \cdots < t_n$ and $A \in \mathcal{B}(\mathbb{R}^n)$ set, with $t_0 := 0$, $x_0 := 0$,

$$\mu_{t_1 \dots t_n}(A) := \int \cdots \int \mathbb{1}_A(x_1, \dots, x_n) \prod_{i=1}^{n} p_{t_i - t_{i-1}}(x_{i-1}, x_i)\,dx_n \cdots dx_1$$

(if $t_1 = 0$, the first factor means $\delta_0(dx_1)$). This is a probability measure: the integrand is a nonnegative Borel density (Theorem 1.47), and integrating out $x_n, x_{n-1}, \dots$ successively gives total mass 1. By Fubini and Theorem 1.9 it is the unique probability measure with

$$\mu_{t_1 \dots t_n}(A_1 \times \cdots \times A_n) = \int_{A_1} \cdots \int_{A_n} \prod_{i=1}^{n} p_{t_i - t_{i-1}}(x_{i-1}, x_i)\,dx_n \cdots dx_1. \tag{4.6}$$

(Replacing $x_0 = 0$ by $x_0 = x$ gives Brownian motion started at x.) For unordered times, sort them by the unique permutation σ with $t_{\sigma(1)} < \cdots < t_{\sigma(n)}$ and permute the coordinates accordingly. The family is consistent: (4.3) holds by construction, and (4.4) amounts to deleting one time,

$$\mu_{t_1 \dots t_{i-1} t_{i+1} \dots t_n}(\cdots \times A_{i-1} \times A_{i+1} \times \dots) = \mu_{t_1 \dots t_n}(\cdots \times A_{i-1} \times \mathbb{R} \times A_{i+1} \times \dots). \tag{4.7}$$

For $i = n$ this is immediate. For $i < n$ it is the **Chapman–Kolmogorov equation**

$$\int_A p_{t+s}(x,z)\,dz = \int_A \int_{\mathbb{R}} p_s(x,y) p_t(y,z)\,dy\,dz, \qquad \text{i.e.} \qquad \int p_s(x,y) p_t(y,z)\,dy = p_{s+t}(x,z),$$

which holds because $N(0,s) * N(0,t) = N(0,s+t)$ (Example 3.44(i)). [Supplement]

The difficulty. By Theorem 4.7 there is a unique $\tilde{P}$ on $(\mathbb{R}^{\mathbb{R}_+}, \sigma(\mathcal{C}_{\mathbb{R}_+}))$ with these marginals, and $X_t(\omega) = \omega(t)$ satisfies Definition 4.1(i)–(iii) (see below). But (iv) fails to make sense: $C(\mathbb{R}_+) \notin \sigma(\mathcal{C}_{\mathbb{R}_+})$. Sets in $\sigma(\mathcal{C}_{\mathbb{R}_+})$ depend on only countably many coordinates, and continuity cannot be decided from countably many values. So we first restrict to the dyadic rationals $\mathbb{Q}_2^+$: the restricted family is still consistent, and Theorem 4.7 gives a unique Q on $(\mathbb{R}^{\mathbb{Q}_2^+}, \sigma(\mathcal{C}_{\mathbb{Q}_2^+}))$ with these marginals for $X_t(\omega) = \omega(t)$, $t \in \mathbb{Q}_2^+$.

Theorem 4.8

Let $C(\mathbb{Q}_2^+) := \bigcap_{n\ge1}\{\omega \in \mathbb{R}^{\mathbb{Q}_2^+} \mid \omega|_{[0,n]\cap\mathbb{Q}_2^+} \text{ is uniformly continuous}\}$. Then $C(\mathbb{Q}_2^+) \in \sigma(\mathcal{C}_{\mathbb{Q}_2^+})$ and $Q(C(\mathbb{Q}_2^+)) = 1$; so $(C(\mathbb{Q}_2^+), C(\mathbb{Q}_2^+) \cap \sigma(\mathcal{C}_{\mathbb{Q}_2^+}), Q)$ is a probability space (with the trace σ-algebra).

From Theorem 4.8 to Brownian motion. Let $\Omega := C(\mathbb{R}_+)$ with the metric

$$\rho(\omega,\omega') := \sum_{n\ge1} 2^{-n}\Big(1 \wedge \sup_{t\in[0,n]} |\omega(t)-\omega'(t)|\Big),$$

under which Ω is a complete separable metric space; $\mathcal{F} := \mathcal{B}(\Omega)$ and $B_t(\omega) := \omega(t)$, which is ρ-continuous and hence measurable, so $\sigma(B_t \mid t \ge 0) \subset \mathcal{F}$. Conversely, for a countable dense $D \subset \Omega$ and $\omega_0 \in D$, $\rho(\omega,\omega_0) = \sum_n 2^{-n}(1 \wedge \sup_{t\in[0,n]\cap\mathbb{Q}_2^+} |B_t(\omega)-\omega_0(t)|)$ is $\sigma(B_t \mid t \in \mathbb{Q}_2^+)$-measurable. Every open set is a countable union of balls $B_\rho(\omega_0, r)$, $\omega_0 \in D$, $r \in \mathbb{Q}_2^+$, so $\mathcal{F} = \sigma(B_t \mid t \ge 0)$.

The map $\psi : C(\mathbb{Q}_2^+) \to \Omega$, $\omega \mapsto$ the unique continuous extension of ω to $\mathbb{R}_+$, is measurable: for $q_m \to t$ in $\mathbb{Q}_2^+$,

$$\psi^{-1}(\{B_t < \alpha\}) = \bigcup_k \bigcup_n \bigcap_{m\ge n} \{\omega \mid \omega(q_m) < \alpha - \tfrac{1}{k}\},$$

and the sets $\{B_t < \alpha\}$ generate $\mathcal{F}$. Let $P := Q \circ \psi^{-1}$. For dyadic t_i, $(\psi\omega)(t_i) = \omega(t_i)$, so

$$P((B_{t_1},\dots,B_{t_n}) \in A_1\times\cdots\times A_n) = Q((X_{t_1},\dots,X_{t_n}) \in A_1\times\cdots\times A_n) \qquad (t_i \in \mathbb{Q}_2^+). \tag{4.8}$$

Then $B_t(\omega) = \omega(t)$ is a Brownian motion on $(\Omega,\mathcal{F},P)$, every path being continuous. The key is that for all $0 \le t_1 < \cdots < t_n$ in $\mathbb{R}_+$, with $t_0 = 0$, $x_0 = 0$,

$$P((B_{t_1},\dots,B_{t_n}) \in A_1\times\cdots\times A_n) = \int_{A_1}\cdots\int_{A_n} \prod_{i=1}^n p_{t_i-t_{i-1}}(x_{i-1},x_i)\,dx_n\cdots dx_1, \tag{4.9}$$

and that (4.9) is equivalent to Definition 4.1(i)–(iii).

[Supplement] *Proof of* (4.9). Take dyadic $t_i^k \to t_i$ (strictly ordered). For $f \in C_b(\mathbb{R}^n)$, $\mathbb{E}_P[f(B_{t_1^k},\dots,B_{t_n^k})] = \int f\,d\mu_{t_1^k\dots t_n^k}$ by (4.8). As $k\to\infty$ the left side tends to $\mathbb{E}_P[f(B_{t_1},\dots,B_{t_n})]$ by path continuity and the DCT. In the right side substitute $x_i = x_{i-1} + \sqrt{t_i^k - t_{i-1}^k}\,z_i$: it becomes an integral of f evaluated at points depending continuously on k against a fixed Gaussian density, so the DCT gives the limit $\int f\,d\mu_{t_1\dots t_n}$. By Example 2.30 on $\mathbb{R}^n$ the two laws agree.

(4.9) *gives (i)–(iii).* $n=1$ gives $B_t \sim N(0,t)$ and $B_0 = 0$ a.s. For $t_1 < t_2$, with $\bar{A} = \{(x,y) \mid y-x \le \alpha\}$ and $z = y-x$,

$$P(B_{t_2}-B_{t_1} \le \alpha) = \int_{\mathbb{R}} p_{t_1}(0,x)\int_{-\infty}^{x+\alpha} p_{t_2-t_1}(x,y)\,dy\,dx = \int_{-\infty}^{\alpha} \frac{e^{-z^2/(2(t_2-t_1))}}{\sqrt{2\pi(t_2-t_1)}}dz.$$

More generally, the change of variables $z_i = x_i - x_{i-1}$ turns the density in (4.9) into $\prod_i p_{t_i-t_{i-1}}(0,z_i)$, a product density. So $(B_{t_1}, B_{t_2}-B_{t_1},\dots,B_{t_n}-B_{t_{n-1}})$ has independent $N(0,t_i-t_{i-1})$ coordinates (Theorem 3.42), i.e.

$$P(B_{t_1}\le\alpha_1, B_{t_2}-B_{t_1}\le\alpha_2,\dots,B_{t_n}-B_{t_{n-1}}\le\alpha_n) = P(B_{t_1}\le\alpha_1)\cdots P(B_{t_n}-B_{t_{n-1}}\le\alpha_n). \tag{4.10}$$

Conversely (i)–(iii) determine the joint law of the increments, hence (4.9).

Proof of Theorem 4.8.

Step 1 (measurability). With $\Omega_n := \{\omega \mid \omega|_{[0,n]\cap\mathbb{Q}_2^+} \text{ uniformly continuous}\}$,

$$\Omega_n = \bigcap_{M\ge1}\bigcup_{N\ge1}\bigcap_{t,s\in\mathbb{Q}_2^+\cap[0,n],\ |t-s|\le\frac{1}{N}} \Big\{|X_t - X_s| \le \frac{1}{M}\Big\} \in \sigma(\mathcal{C}_{\mathbb{Q}_2^+}),$$

a countable combination of cylinder sets. It remains to show $Q(\Omega_n) = 1$ for each n.

Step 2 (moments). The marginals of $(X_t)_{t\in\mathbb{Q}_2^+}$ satisfy Definition 4.1(i)–(iii), hence scaling invariance: $X_t - X_s \sim \frac{1}{\sqrt{\lambda}}(X_{\lambda t} - X_{\lambda s}) \sim N(0, t-s)$. For $Y_t^n := \frac{1}{\sqrt{n}} X_{nt}$ and $s < t$ in $\mathbb{Q}_2^+ \cap [0,1]$,

$$\mathbb{E}_Q|Y_t^n - Y_s^n|^{2k} = \mathbb{E}_Q|X_t - X_s|^{2k} = \mathbb{E}_Q|\sqrt{t-s}\, X_1|^{2k} = c_k|t-s|^k, \qquad c_k := \mathbb{E}_Q|X_1|^{2k}. \tag{4.11}$$

Step 3. By Theorem 4.9 and Remark 4.10 there is Ω^* with $Q(\Omega^*) = 1$ on which $t \mapsto Y_t^n(\omega)$ is Hölder continuous of order $\gamma \in (0, \frac{1}{2})$ on $[0,1] \cap \mathbb{Q}_2^+$, i.e. $t \mapsto X_t(\omega)$ is Hölder on $[0,n] \cap \mathbb{Q}_2^+$. So $\Omega^* \subset \Omega_n$. ∎

Theorem 4.9 (Kolmogorov–Čentsov, dyadic version)

Let $(X_t)_{t\in\mathbb{Q}_2^+}$ be a process on $(\mathbb{R}^{\mathbb{Q}_2^+}, \sigma(\mathcal{C}_{\mathbb{Q}_2^+}), Q)$ with

$$\mathbb{E}_Q|X_t - X_s|^\alpha \le K|t-s|^{1+\beta} \qquad \forall t, s \in \mathbb{Q}_2^+ \cap [0,1]$$

for some $\alpha, \beta, K > 0$, and let $\gamma \in (0, \beta/\alpha)$. Then there is $\Omega^* \in \sigma(\mathcal{C}_{\mathbb{Q}_2^+})$ with $Q(\Omega^*) = 1$ such that for every $\omega \in \Omega^*$ there is $n^*(\omega)$ with

$$\sup_{t,s\in\mathbb{Q}_2^+\cap[0,1],\ 0<t-s<2^{-n^*(\omega)}} \frac{|X_t(\omega) - X_s(\omega)|}{|t-s|^\gamma} \le C := \frac{2}{1-2^{-\gamma}}.$$

In particular every $\omega \in \Omega^*$ is Hölder continuous of order γ on $\mathbb{Q}_2^+ \cap [0,1]$: with $n_0 := n^*(\omega)+1$, splitting $[s,t]$ into 2^{n_0} equal pieces gives $|X_t(\omega) - X_s(\omega)| \le 2^{n_0} C\big((t-s)2^{-n_0}\big)^\gamma = 2^{n_0(1-\gamma)} C|t-s|^\gamma$.

Remark 4.10

By (4.11) we may take $\alpha = 2k$, $\beta = k-1$, and $\beta/\alpha = \frac{k-1}{2k} \nearrow \frac{1}{2}$. So Theorem 4.9 applies to every $\gamma \in (0, \frac{1}{2})$: Q-a.e. path is locally Hölder continuous of every order $\gamma < \frac{1}{2}$. The map ψ preserves local Hölder continuity, so P-a.e. Brownian path in $C(\mathbb{R}_+)$ is locally γ-Hölder for every $\gamma < \frac{1}{2}$.

Proof of Theorem 4.9.

Step 1 (Borel–Cantelli on dyadic increments). By Markov's inequality and the hypothesis,

$$\begin{aligned} Q\Big(\max_{1\le k\le 2^n} |X_{k2^{-n}} - X_{(k-1)2^{-n}}| \ge 2^{-\gamma n}\Big) &\le \sum_{k=1}^{2^n} \frac{\mathbb{E}_Q|X_{k2^{-n}} - X_{(k-1)2^{-n}}|^\alpha}{2^{-\gamma n\alpha}} \\ &\le 2^n \frac{K 2^{-n(1+\beta)}}{2^{-\gamma n\alpha}} = K 2^{-n(\beta-\gamma\alpha)}. \end{aligned}$$

This is summable since $\beta > \gamma\alpha$. By Borel–Cantelli, $\Omega^* := \bigcup_l \bigcap_{n\ge l} \{\max_k |X_{k2^{-n}} - X_{(k-1)2^{-n}}| < 2^{-\gamma n}\}$ has $Q(\Omega^*) = 1$, and for $\omega \in \Omega^*$ there is $n^*(\omega)$ with

$$\max_{1\le k\le 2^n} |X_{k2^{-n}}(\omega) - X_{(k-1)2^{-n}}(\omega)| < 2^{-\gamma n} \qquad \forall n \ge n^*(\omega). \tag{4.12}$$

Step 2 (chaining). Fix $\omega \in \Omega^*$, $n \ge n^*(\omega)$, and let $D_m := \{k2^{-m} \mid 0 \le k \le 2^m\}$. We claim, for all $m > n$,

$$|X_t(\omega) - X_s(\omega)| \le 2 \sum_{j=n+1}^{m} 2^{-\gamma j} \qquad \forall t, s \in D_m,\ 0 < t-s < 2^{-n}. \tag{4.13}$$

For $m = n+1$, t, s are neighbours in D_{n+1}, and (4.12) gives $< 2^{-\gamma(n+1)}$. Assume (4.13) for $m = M-1$, and let $s, t \in D_M$, $0 < t - s < 2^{-n}$. Put $s_1 := \min\{r \in D_{M-1} \mid r \geq s\}$ and $t_1 := \max\{r \in D_{M-1} \mid r \leq t\}$. Then $s \leq s_1 \leq t_1 \leq t$ and $s_1 - s, t - t_1 \in \{0, 2^{-M}\}$, so (4.12) bounds $|X_{s_1} - X_s|$ and $|X_t - X_{t_1}|$ by $2^{-\gamma M}$, and the induction hypothesis bounds $|X_{t_1} - X_{s_1}|$. Summing gives (4.13) for $m = M$.

Step 3. For $t, s \in \mathbb{Q}_2^+ \cap [0,1]$ with $0 < t - s < 2^{-n^*(\omega)}$ choose $m > n \geq n^*(\omega)$ with $t, s \in D_m$ and $2^{-(n+1)} \leq t - s < 2^{-n}$. By (4.13),

$$|X_t - X_s| \leq 2 \sum_{j \geq n+1} 2^{-\gamma j} = \frac{2 \cdot 2^{-\gamma(n+1)}}{1 - 2^{-\gamma}} \leq \frac{2}{1 - 2^{-\gamma}} |t - s|^{\gamma}.$$

■

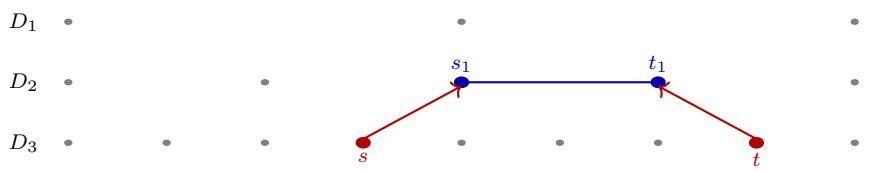


Figure. *Chaining in the proof of Theorem 4.9: $s, t \in D_M$ are moved by at most one step of size 2^{-M} to $s_1, t_1 \in D_{M-1}$, and the induction continues on coarser grids. The increments along the chain are bounded by $2^{-\gamma j}$ at level j, a convergent geometric series.*

Remark 4.11

$((B_t)_{t \geq 0}, \Omega, \mathcal{F}, P)$ as constructed above is the **canonical model** of Brownian motion; $P = P_0$ is the **Wiener measure** (started at 0).

4.3 The Markov property

Motivation. Brownian motion forgets its past: given the present position B_s, the future path $(B_{s+t})_{t \geq 0}$ is a Brownian motion started at B_s, regardless of how it got there. On the canonical space this is expressed with the shift θ_s. Working with the slightly larger filtration $\mathcal{F}_t^{0,+}$ costs nothing and yields Blumenthal's 0–1 law.

Work in the canonical model: $\Omega = C(\mathbb{R}_+)$, $B_t(\omega) = \omega(t)$, $\mathcal{F} = \mathcal{B}(C(\mathbb{R}_+)) = \sigma(B_t \mid t \geq 0)$. For a start in $x \in \mathbb{R}$ there are three equivalent options: (1) (B_t) under P_x, built as P but with $x_0 = x$ in (4.6); (2) $(B_t + x)$ under P_0; (3) (B_t) under $P_0 \circ T_x^{-1}$ with $T_x(\omega) := \omega + x$. All have the same finite-dimensional distributions and continuous paths; in particular $P_x = P_0 \circ T_x^{-1}$. We write $\mathbb{E}_x[Y] := \int Y \, dP_x$.

Let $\mathcal{F}_t^0 := \sigma(B_u \mid u \leq t)$. Then $(\mathcal{F}_t^0)_{t \geq 0}$ is a **filtration** ($\mathcal{F}_s^0 \subset \mathcal{F}_t^0 \subset \mathcal{F}$ for $s \leq t$), the "information up to time t", and (B_t) is **adapted** (B_t is $\mathcal{F}_t^0$-measurable). The **shift** $\theta_s : \Omega \to \Omega$, $\theta_s(\omega)(t) := \omega(s+t)$, is ρ-continuous, hence measurable, and in fact $\hat{\mathcal{F}}_s^0/\mathcal{F}$-measurable with $\hat{\mathcal{F}}_s^0 := \sigma(B_u \mid u \geq s)$, the future after s. For $Y \in \mathcal{F}_b$ (bounded $\mathcal{F}$-measurable), $Y \circ \theta_s$ is "a function of the future after time s". For instance, $Y = f(B_{t_1}, \ldots, B_{t_n})$ gives $Y \circ \theta_s = f(B_{t_1+s}, \ldots, B_{t_n+s})$, and $B_t \circ \theta_s = B_{t+s}$.

The canonical model has the **Markov property** w.r.t. $(\mathcal{F}_t^0)$: for $Y \in \mathcal{F}_b$,

$$\mathbb{E}_x[\underbrace{Y \circ \theta_s}_{\text{future}} \mid \underbrace{\mathcal{F}_s^0}_{\text{past}}] = \mathbb{E}_{\underbrace{B_s}_{\text{present}}}[Y] \qquad P_x\text{-a.s., } \forall s \geq 0,\ x \in \mathbb{R}. \tag{4.14}$$

It will follow from the Markov property (4.15) for a larger filtration (Remark 4.13).

Remark 4.12

$x \mapsto \mathbb{E}_x[Y]$ is Borel measurable. [Supplement] *Reason:* the class of $Y \in \mathcal{F}_b$ for which this holds is a monotone vector space (MCT). It contains the multiplicative class of indicators of cylinder sets, for which $\mathbb{E}_x[\mathbb{1}_A]$ is an integral of Gaussian kernels continuous in x, by (4.6) and the DCT. Theorem 2.29 applies. Hence $\mathbb{E}_{B_s}[Y]$ is $\sigma(B_s)$-measurable, and since $\sigma(B_s) \subset \mathcal{F}_s^0$, P_x-a.s.

$$\begin{aligned}\mathbb{E}_x[Y \circ \theta_s \mid \sigma(B_s)] &= \mathbb{E}_x\big[\mathbb{E}_x[Y \circ \theta_s \mid \mathcal{F}_s^0] \,\big|\, \sigma(B_s)\big] = \mathbb{E}_x[\mathbb{E}_{B_s}[Y] \mid \sigma(B_s)] \\ &= \mathbb{E}_{B_s}[Y] = \mathbb{E}_x[Y \circ \theta_s \mid \mathcal{F}_s^0],\end{aligned}$$

using Theorem 2.35(iv), (4.14) and Example 2.34(i). The best guess for the future given the present equals the best guess given the whole past.

For technical reasons we enlarge $\mathcal{F}_t^0$ to

$$\mathcal{F}_t^{0,+} := \bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}^0 \ (\supset \mathcal{F}_t^0),$$

which allows an infinitesimal look into the future. It is **right continuous**: $\bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}^{0,+} = \bigcap_n \bigcap_k \mathcal{F}_{t+\frac{1}{n}+\frac{1}{k}}^0 \subset \mathcal{F}_t^{0,+}$, and trivially $\mathcal{F}_t^{0,+} \subset \bigcap_{\varepsilon>0} \mathcal{F}_{t+\varepsilon}^{0,+}$. (B_t) is adapted to $(\mathcal{F}_t^{0,+})$ and has the Markov property w.r.t. it:

$$\mathbb{E}_x[Y \circ \theta_s \mid \mathcal{F}_s^{0,+}] = \mathbb{E}_{B_s}[Y] \qquad P_x\text{-a.s., } \forall s \geq 0,\ x \in \mathbb{R},\ Y \in \mathcal{F}_b. \tag{4.15}$$

Proof of (4.15).

Step 1 (reduction). Let $\mathcal{C}$ be the cylinder sets $A = \{(B_{t_1}, \dots, B_{t_n}) \in A_1 \times \cdots \times A_n\}$, $0 < t_1 < \cdots < t_n$, $A_i \in \mathcal{B}(\mathbb{R})$; $\mathcal{C}$ is $\cap$-stable and generates $\mathcal{F}$. $\mathcal{H} := \{Y \in \mathcal{F}_b \mid (4.15) \text{ holds}\}$ is an MVS (conditional MCT), so by Theorem 2.29 it suffices to treat $Y = \mathbb{1}_A$, $A \in \mathcal{C}$, i.e. to show

$$\mathbb{E}_x[\mathbb{1}_A \circ \theta_s \, \mathbb{1}_C] = \mathbb{E}_x[\mathbb{E}_{B_s}[\mathbb{1}_A] \, \mathbb{1}_C] \qquad \forall C \in \mathcal{F}_s^{0,+}. \tag{4.16}$$

Step 2 (special C). Fix x, s and $0 < h < t_1$, and take $C = \{(B_{s_1}, \dots, B_{s_m}) \in C_1 \times \cdots \times C_m\}$ with $0 \leq s_1 < \cdots < s_m = s + h$. With $s_0 := 0$, $y_0 := x$, (4.9) gives

$$\mathbb{E}_x[\mathbb{1}_A \circ \theta_s \, \mathbb{1}_C] = \int_{C_1} \cdots \int_{C_m} \prod_{j=1}^{m} p_{s_j - s_{j-1}}(y_{j-1}, y_j) \, F(y_m, h) \, dy_m \cdots dy_1 = \mathbb{E}_x[F(B_{s+h}, h) \mathbb{1}_C], \tag{4.17}$$

where, with $x_0 := y$ and $t_0 := h$,

$$F(y, h) := \int_{A_1} \cdots \int_{A_n} p_{t_1 - h}(y, x_1) \prod_{i=2}^{n} p_{t_i - t_{i-1}}(x_{i-1}, x_i) \, dx_n \cdots dx_1.$$

Step 3 (all $C \in \mathcal{F}_{s+h}^0$). The sets C of Step 2 form a $\cap$-stable generator of $\mathcal{F}_{s+h}^0$, and the class of $\tilde{C} \in \mathcal{F}_{s+h}^0$ satisfying (4.17) is a Dynkin system. By Theorem 1.8,

$$\mathbb{E}_x[\mathbb{1}_A \circ \theta_s \, \mathbb{1}_C] = \mathbb{E}_x[F(B_{s+h}, h) \mathbb{1}_C] \qquad \forall C \in \mathcal{F}_{s+h}^0 \supset \mathcal{F}_s^{0,+}. \tag{4.18}$$

Step 4 ($h \to 0$). By the DCT, F is jointly continuous in $(y, h) \in \mathbb{R} \times (0, t_1)$ and bounded by 1. As $h \to 0$, path continuity gives $F(B_{s+h}(\omega), h) \to \int_{A_1} \cdots \int_{A_n} p_{t_1}(B_s(\omega), x_1) \prod_{i \geq 2} p_{t_i - t_{i-1}}(x_{i-1}, x_i) \, dx = \mathbb{E}_{B_s(\omega)}[\mathbb{1}_A]$. The DCT in (4.18) then yields (4.16) for all $C \in \mathcal{F}_s^{0,+}$. ∎

Remark 4.13

(4.15) implies (4.14): $\mathbb{E}_x[Y\circ\theta_s \mid \mathcal{F}_s^0] = \mathbb{E}_x[\mathbb{E}_x[Y\circ\theta_s \mid \mathcal{F}_s^{0,+}] \mid \mathcal{F}_s^0] = \mathbb{E}_x[\mathbb{E}_{B_s}[Y] \mid \mathcal{F}_s^0] = \mathbb{E}_{B_s}[Y]$ by Theorem 2.35(iv), (4.15) and Example 2.34(i).

Theorem 4.14

For $Z \in \mathcal{F}_b$: $\mathbb{E}_x[Z \mid \mathcal{F}_s^{0,+}] = \mathbb{E}_x[Z \mid \mathcal{F}_s^0]$ P_x-a.s., for all $s \geq 0$, $x \in \mathbb{R}$.

Proof. By the monotone class theorem it suffices to take $Z = \mathbb{1}_A$ with $A = \{(B_{t_1}, \dots, B_{t_n}) \in A_1 \times \cdots \times A_n\}$, $0 \leq t_1 < \cdots < t_n$. Then, P_x-a.s., by Theorem 2.35(iii) and (4.14), (4.15) (both equal $\mathbb{E}_{B_s}[\cdot]$),

$$\begin{aligned}\mathbb{E}_x[\mathbb{1}_A \mid \mathcal{F}_s^{0,+}] &= \prod_{t_i\leq s} \mathbb{1}_{A_i}(B_{t_i})\, \mathbb{E}_x\Big[\prod_{t_i>s} \mathbb{1}_{A_i}(B_{t_i-s})\circ\theta_s \,\Big|\, \mathcal{F}_s^{0,+}\Big] \\ &= \prod_{t_i\leq s} \mathbb{1}_{A_i}(B_{t_i})\, \mathbb{E}_x\Big[\prod_{t_i>s} \mathbb{1}_{A_i}(B_{t_i-s})\circ\theta_s \,\Big|\, \mathcal{F}_s^{0}\Big] = \mathbb{E}_x[\mathbb{1}_A \mid \mathcal{F}_s^0].\end{aligned}$$

■

Remark 4.15

$\mathcal{F}_s^{0,+}$ and $\mathcal{F}_s^0$ differ only by P_x-null sets (for every x). For $Z \in \mathcal{F}_s^{0,+}$ bounded, $Z = \mathbb{E}_x[Z \mid \mathcal{F}_s^{0,+}] = \mathbb{E}_x[Z \mid \mathcal{F}_s^0]$ P_x-a.s., an $\mathcal{F}_s^0$-measurable r.v. With $Z = \mathbb{1}_{B^c}$, $B \in \mathcal{F}_s^{0,+}$, the set $B = \{\mathbb{1}_{B^c} < 1\}$ agrees up to a P_x-null set with $B_0 := \{\mathbb{E}_x[\mathbb{1}_{B^c} \mid \mathcal{F}_s^0] < 1\} \in \mathcal{F}_s^0$.

Theorem 4.16 (Blumenthal's 0–1 law)

If $A \in \mathcal{F}_0^{0,+}$, then $P_x(A) \in \{0,1\}$ for every $x \in \mathbb{R}$.

Proof. $\mathcal{F}_0^0 = \sigma(B_0)$ and $P_x(B_0 = x) = 1$, so every $\mathcal{F}_0^0$-measurable Y is P_x-a.s. constant, $Y = \mathbb{E}_x[Y]$. For $A \in \mathcal{F}_0^{0,+}$, by Example 2.34(i) and Theorem 4.14, P_x-a.s.

$$\mathbb{1}_A = \mathbb{E}_x[\mathbb{1}_A \mid \mathcal{F}_0^{0,+}] = \mathbb{E}_x[\mathbb{1}_A \mid \mathcal{F}_0^0] = \mathbb{E}_x[\mathbb{E}_x[\mathbb{1}_A \mid \mathcal{F}_0^0]] = P_x(A),$$

so $P_x(A) \in \{0,1\}$. ■

Remark 4.17

A general BM as in Definition 4.1 (started at x) also has the Markov property w.r.t. $(\mathcal{F}_t^{0,+})$. Since a shift need not exist, it is formulated as

$$\mathbb{E}_x\Big[\prod_{i=1}^n \mathbb{1}_{A_i}(B_{t_i+s}) \,\Big|\, \mathcal{F}_s^{0,+}\Big] = \mathbb{E}_{B_s}\Big[\prod_{i=1}^n \mathbb{1}_{A_i}(B_{t_i})\Big] \qquad P_x\text{-a.s.}, \tag{4.19}$$

for cylinder sets as above. The proof of (4.16) used only the finite-dimensional distributions (4.9) and a.s. path continuity. A general BM satisfies (4.9) (equivalent to Definition 4.1(i)–(iii), also for start in x), so it has the same finite-dimensional distributions as the canonical one. In particular $\mathbb{E}_x[f(B_{t+s}) \mid \mathcal{F}_s^{0,+}] = \mathbb{E}_{B_s}[f(B_t)]$ for $f \in \mathcal{B}(\mathbb{R})_b$.

4.4 The d-dimensional Brownian motion

Motivation. In $\mathbb{R}^d$, Brownian motion is simply d independent one-dimensional Brownian motions as coordinates. The Gaussian kernel becomes d-dimensional, and the constructions and the Markov property carry over verbatim.

Let $\|\cdot\|$ be the Euclidean norm and

$$p_t(x,y) := \frac{1}{(\sqrt{2\pi t})^d} e^{-\frac{\|y-x\|^2}{2t}} \quad (x,y\in\mathbb{R}^d,\ t>0), \qquad p_0(x,y)\,dy := \delta_x(dy). \tag{4.20}$$

The construction of Section 4.2 goes through with this kernel; Kolmogorov's extension theorem holds for $\mathbb{R}^d$-valued (indeed Polish-space-valued) processes, with the same proof (see also Le Gall, *Brownian Motion, Martingales, and Stochastic Calculus*, Thm. 6.3). Equivalently: if P_0 is the one-dimensional Wiener measure on $(C(\mathbb{R}_+),\mathcal{B}(C(\mathbb{R}_+)))$ and $P^{(i)} = P_0$, $i=1,\dots,d$, then $\mathbf{P}_0 := \bigotimes_{i=1}^d P^{(i)}$ is the **d-dimensional Wiener measure** on $\Omega := C(\mathbb{R}_+)^d = C(\mathbb{R}_+,\mathbb{R}^d)$, $\mathcal{F} := \mathcal{B}(C(\mathbb{R}_+)^d)$. Under $\mathbf{P}_0$, $W_t(\omega) := \omega(t) = (\omega_1(t),\dots,\omega_d(t))$ is the canonical d-dimensional BM from 0, and under $P_x := \mathbf{P}_0\circ\Pi_x^{-1}$, $\Pi_x(\omega)=\omega+x$, it starts at x. With $\mathcal{F}_t^0 := \sigma(W_s \mid s\le t)$ and $\mathcal{F}_t^{0,+} := \bigcap_{\varepsilon>0}\mathcal{F}_{t+\varepsilon}^0$ (again filtrations to which W is adapted),

$$P_x((W_{t_1},\dots,W_{t_n})\in A_1\times\cdots\times A_n) = \int_{A_1}\cdots\int_{A_n}\prod_{i=1}^n p_{t_i-t_{i-1}}(x_{i-1},x_i)\,dx_1\cdots dx_n \tag{4.21}$$

for $0\le t_1<\cdots<t_n$, $A_i\in\mathcal{B}(\mathbb{R}^d)$, $x_0=x$, $t_0=0$, with d-dimensional Lebesgue measure dx_i. The analogues of (4.14), (4.15), Theorem 4.14, Remark 4.15 and Theorem 4.16 hold with essentially the same proofs. The components W^i are independent BMs: by (4.21), which factorises over coordinates, $P_x(\bigcap_{i=1}^d \tilde A_i) = \prod_i P_x(\tilde A_i)$ for cylinder sets $\tilde A_i = \{(W^i_{t_1},\dots,W^i_{t_n})\in B^i_1\times\cdots\times B^i_n\}$, and Theorem 3.10 extends this to the generated σ-algebras.

Definition 4.18 (d-dimensional Brownian motion)

An $\mathbb{R}^d$-valued process $(W_t)_{t\ge0}$ on some $(\Omega,\mathcal{F},P_x)$ is a **d-dimensional BM started at** x if (i) $P_x(W_0=x)=1$; (ii) for $0\le s<t$, $W_t-W_s\sim N(0,(t-s)I_d)$, i.e. $P_x(W_t-W_s\in A)=\int_A p_{t-s}(0,y)\,dy$; (iii) the increments $W_{t_1}-W_0,\dots,W_{t_n}-W_{t_{n-1}}$ are independent for $0\le t_1<\cdots<t_n$; (iv) almost all paths are continuous.

Remark 4.19

A d-dimensional BM satisfies (4.21), which is equivalent to Definition 4.18(i)–(iii); so it has the finite-dimensional distributions of the canonical one. For it only P_x-a.e. path is continuous, whereas for the canonical model every path is. As in Remark 4.17 it has the Markov property

$$\mathbb{E}_x\Big[\prod_{i=1}^n \mathbb{1}_{A_i}(W_{t_i+s})\,\Big|\,\mathcal{F}_s^{0,+}\Big] = \mathbb{E}_{W_s}\Big[\prod_{i=1}^n \mathbb{1}_{A_i}(W_{t_i})\Big] \quad P_x\text{-a.s.}, \tag{4.22}$$

and in particular $\mathbb{E}_x[f(W_{t+s})\mid\mathcal{F}_s^{0,+}] = \mathbb{E}_{W_s}[f(W_t)]$ for $f\in\mathcal{B}(\mathbb{R}^d)_b$.